

EDUCATION

- **M.Sc. Computer Science**, Georgia Institute of Technology, GA. 2024 - 2026
- **B.Sc. Computer Science and Engineering**, Yonsei University, South Korea. 2016 - 2018, 2021 - 2023

TECHNOLOGIES

- Technologies: Python, C++, Java, C#, PyTorch, Cloud Computing (AWS, Firebase), CMake, MATLAB, Web Scraping (Scrapy, Playwright), Docker, Express.js, SQL, NoSQL (MongoDB, DynamoDB), React JS, FastAPI, Unity 3D
- Other: NLP, ML/DL, AI Agents, Virtualization, Linux Server Administration

WORK EXPERIENCE

Software Dev Engineer Intern Amazon Web Services May 2025 - Aug 2025
Seattle WA, USA

- Implemented efficient cross-database reconciliation algorithm eliminating linear memory scaling, for internal AWS service managing network rules of elastic compute instances.
Observing significant reduction in memory usage and garbage collection overhead, new components pushed to production after implementing and running a suite of tests and performance benchmarks.
- Utilized Amazon CloudWatch to analyse and aggregate logs and produce data visualizations to compare latency and compute (memory/CPU usage & GC statistics).

Software Developer & ML Engineer MeTown 2023 - Aug 2024
Seoul, South Korea

- Developed clothes size recommendation system, utilizing customer purchase data to train and test machine learning models, deployed as a REST API.
My algorithm increased the accuracy of size prediction by 10% over the previous algorithm. It was served to an online clothing store's backend service to reduce manual size recommendation labour, and it led to fewer refunds due to size mismatch.
- Developed microservices in Python and Express with JWT-based authentication, and used Firebase to track API usage.
- Developed an [asynchronous web scraper](#) for clothing size data and customer reviews.

Software Developer (Part Time) Center for High Throughput Computing Mar 2022 - May 2022
Madison WI, USA

- Worked on HTCondor, a large open-source distributed high throughput computing system written in C++.
- Updated legacy libraries in CMake, and applied C++ 20 compile time optimizations for static configs.

Software Developer KOG 2018 - 2020
Daegu, South Korea

- Developed GUI application for light simulation in C++. Developed image filters, conversions between color spaces, and various UI elements to help visualize simulation results involving lights.
- Developed AR application in Unity for a handheld medical device, projecting virtual screens on a moving object by tracking special markers.

PROJECTS

- [Godot Engine \(C++ Game Engine\)](#): Fixed a bug in C# scripting engine related to scene inheritance ([Merged PR](#)).
- [audiobookutils](#): Developed an audiobook and e-book synchronization tool utilizing OpenAI's Whisper.
- [Topic Modeling of EEBO-TCP Data](#): Developed a natural language analysis and topic modelling tool for historical texts with Georgia Tech students and faculty.
- [ComfyUI Usability](#): Developed a collection of utilities for ComfyUI, a node-based backend/UI for AI image/video generation.